



Transceiver
ER130

User's Guide

Copyright

© Opsis AB. All rights reserved. This manual and the product described in it are copyrighted with all rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system or translated into any language in any form by any means without the written permission of Opsis AB. Information in this document is subject to change without notice.

Trademarks

OPSIS is a trademark owned by Opsis AB, Sweden. Other brands and product names are trademarks or registered trademarks of their respective companies. Companies, names, and data used in examples herein are fictitious unless otherwise noted.

Limited warranty

The warranty conditions of this product are described in the purchase agreement between the buyer and the seller. However, a condition for any warranty is that the product is installed, operated and maintained in accordance with the specifications and instructions issued by Opsis AB including but not limited to this manual. This warranty is limited to replacement of defective material or documentation supplied by Opsis AB and in no event shall Opsis AB or its suppliers be liable for any loss of profit or any other commercial damage, including but not limited to direct or indirect, special, incidental, contingent or consequential damages resulting from any defects on the product or its documentation including damages from downtime, loss of goodwill, damage to or replacement of equipment or property.



Contact information

Adress	Opsis AB Box 244 SE-244 02 Furulund, Sweden
Telephone	+46 46 72 25 00
Fax	+46 46 72 25 01
Web-site	http://www.opsis.se
E-mail	info@opsis.se

Contents

1	The equipment	1
1.1	The emitter lamp	1
1.2	Technical specifications	2
1.3	The transceiver	3
1.3.1	ER130 lamp holder	4
1.4	The lamp	5
1.4.1	ER130 fibre holder	6
1.5	The Retro-Reflector RR090	7
1.6	The control unit	8
1.6.1	The power supply card (PM0)	10
1.6.2	The stepping motor cards	11
1.6.3	The flip-flop power card	12
1.6.4	Cables	12
1.7	The UV Filter UF225	14
2	Before using the emitter/receiver	15
2.1	Safety regulations	15
2.2	Precautionary measures	16
3	Installation	17
3.1	Optimizing the light	18
3.1.1	Starting from scratch	18
3.1.2	Precision adjustments	20
3.2	Installation procedure	21
4	Maintenance	23
4.1	Cleaning the mirrors and windows	23
4.2	Lamp replacement	23
4.2.1	Procedure for lamp replacement	24
5	Troubleshooting	25

The ER130 unit is a remotely controlled combined transmitter and receiver, a transceiver. It is designed for the measurement of atmospheric gases in outdoor environments, together with an opto-analysis unit from the AR500 series. The monitoring paths are folded, and the far ends are defined by corner cube mirrors, retro-reflectors. When a monitoring path has been installed, the coordinates for the path are stored in the analyser. Each time the unit returns to a certain monitoring path the light level is automatically optimized. Up to 12 paths can be installed.

ER130 should not be used for longer paths than 250 metres, single way, which corresponds to monitoring paths of up to 500 metres. The system is therefore not suitable for background monitoring.

Two types of retro-reflectors are available, one has 7 prisms, the other one has 12. The one with 12 is useful for all monitoring paths, whilst the one with 7 should not be used for longer paths than 150 metres, single way.

As the ER130 system is simple to install, and as electric power is required only at one side of the monitoring path, the system may well be used for temporary monitoring. The container CR300, on which the ER130 transceiver can be installed, is designed for those applications.

The lamp in the transmitter is located at the focal point of a concave mirror. The receiving half is also equipped with a concave mirror. This mirror collects and focuses the received light onto the end of an optic fibre, which is located at the focal point of the mirror.

The transmitter and the receiver are combined into a coaxial configuration, where the optical axis is common.

In order to start and run the system, an Opsi PS150 power supply is required. The power supply is mounted onto the ER130 unit.

1.1 The emitter lamp

The emitter is equipped with a continuous high-pressure xenon lamp, which is located close to the focal point of a parabolic mirror. The light emitted by the emitter is thus almost parallel. The divergence is about 2 mrad.

The standard lamp is designated Type B. This lamp is useful for the monitoring of most components, except nitric oxide and ammonia. When one, or both, of these species are to

be measured the lamp has to be a Type A. Type A is, however, also generating ozone. The monitoring of ozone levels can therefore not be made without an appropriate ventilation of the telescope. For this purpose the fan VF020 should be attached to the emitter telescope.

1.2 Technical specifications

Transceiver ER130	
Material	acid resistant stainless steel ss 2343
Length, incl. cover	1375 mm
Diameter of telescope	205 mm
Height	640 mm
Weight, incl. base plate	approx. 120 kg
Base plate dimensions	500 x 400 mm ²
Lamp mirror	
focal length	approx. 610 mm (24")
diameter of mirror	150 mm
beam divergence	2 mrad
Receiving mirror	
focal length	approx. 450 mm (17.5")
diameter of mirror	76 mm (3")
Window	
material	Suprasil quartz
diameter	150 mm
thickness	5 mm
Angular rotation	
horizontal	±185°
vertical	±15°
Max. delay time between two paths	20 seconds
Working temperature	-30 °C to +50 °C
Degree of protection	IP 54
Recommended monitoring path length	75-250 metres (single way)
Max. number of paths	12

Retroreflector RR090	
Material	stainless steel ss 2343
Length, excl. attachment	350 mm
Diameter	130 mm
Weight	approx. 5.5 kg
Prism	
material	quartz, suprasil
number of prisms	7 (for path lengths 75 to 150 m) or 12 (path lengths 75 to 250 m)

Control unit	
Dimensions (H×W×D)	135×485×310 mm
Weight	approx. 10 kg
Working temperature	+5 °C to 30 °C
Power requirement	230 V _{AC} (+6 %, -10 %) 300 W 115 V _{AC} (±10 %), 300 W
Cable lengths (standard)	2 metres
analyser - control unit	2 metres
control unit - ER130	5 metres

1.3 The transceiver

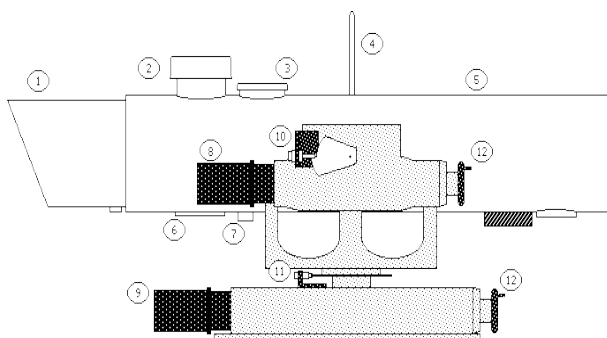


Figure 1.1. Left hand side of the transceiver. The cover plate over the stepping motor is removed.

ER130 transceiver, left side	
1	Protective hood
2	Ventilation for lamp
3	Access opening to fibre holder
4	Lifting eye
5	Radiation shield
6	Lamp holder base plate
7	Access hole for fibre optic cable
8	Stepping motor for vertical movement
9	Stepping motor for horizontal movement
10	Position sensor for vertical home
11	Position sensor for horizontal home
12	Handle for manual movements of the transceiver

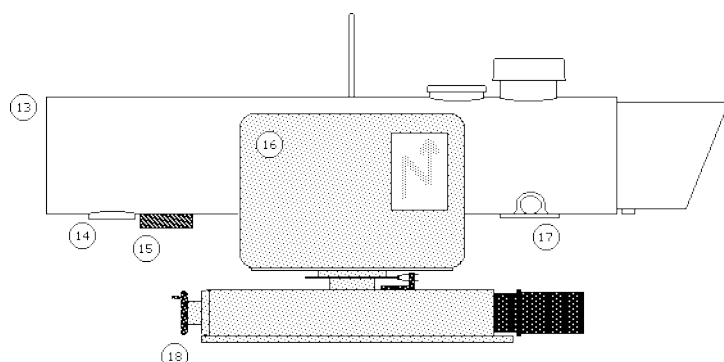


Figure 1.2. The transceiver, the right hand side.

ER130 transceiver, right side	
13	C over plate for the rear mirror
14	Base plate for the smaller mirror
15	Fan (optional)
16	Power supply PS150
17	Access hole to lateral adjustment screw on lamp holder
18	Base plate ER130

1.3.1 ER130 lamp holder

The lamp holder is attached to the telescope from below by four screws. These should normally not have to be loosened when changing the lamp, as all important screws can be reached by removing the quartz-window in the front and through the access-holes.

There are three fine adjustment screws. One is reached from the front (the focusing screw), one is reached through an access hole in the base plate (the vertical screw), and the last one (the horizontal screw) is reached through the telescope side plate (item 17 in [figure 1.2](#)).

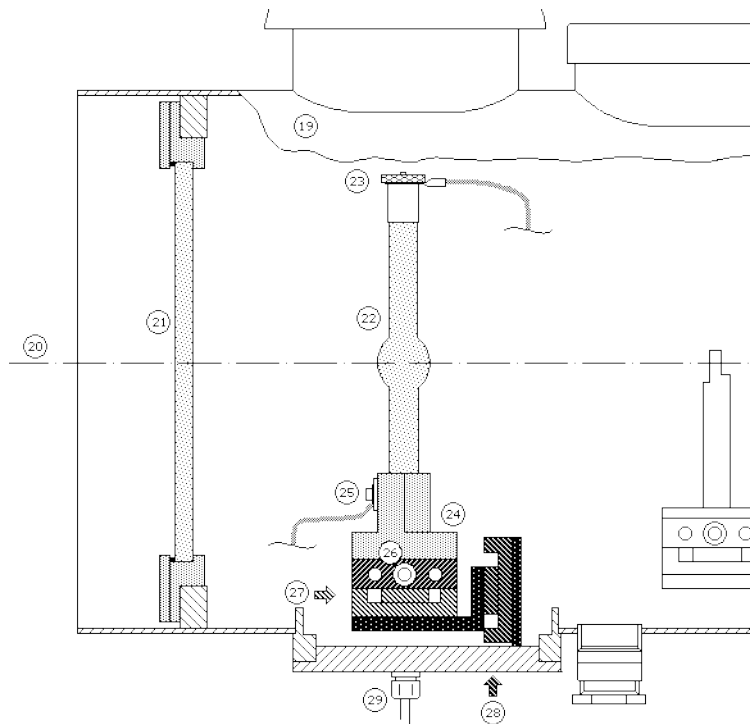


Figure 1.3. Lamp holder, side view.

ER130 lamp holder	
19	Telescope
20	Optical axis
21	Quartz window
22	Xenon lamp
23	Cable-locking nut for power cable (+)
24	Ceramic lamp socket
25	Fixing screws for lamp and power cable (-)
26	Horizontal adjustment screw
27	Focusing screw
28	Vertical adjustment screw
29	Power cable grip nuts.ER130

1.4 The lamp

The xenon lamp used in the emitter is a point light source. Light is generated by spark discharge through high-purity xenon gas, at a pressure of about 30 bar. The lamp is powered by a stable DC supply, and requires a short high-voltage initiation pulse of a few tens of kV. The pressure in the lamp is still about 10 bars when it is cold.

The radiation from the lamp consists of ultraviolet, visible and infrared wavelengths. The wavelength distribution is continuous, apart from some peaks in the visible region, see [section 1.4](#) below.

Two different types of lamp are available. The only difference between them lies in the far ultraviolet part of the spectrum. The Type B lamp has a cut-off at wavelengths shorter than 220 nm. The Type A lamp produces radiation down to 185 nm, but generates ozone. The former can be used for the measurement of most compounds except NO and NH₃. The latter is used in the measurement of NO and NH₃ but cannot be used to measure ozone, if not the ventilation fan VF020 is connected.

The lamps have the following specifications:

Lamp specification	
Lamp type	Type B, ozone-free (standard) Type A, ozone generating
Power dissipation	150 W
Spark length	2.0 mm
Lamp current	8.0 (± 0.5) A _{DC}
Voltage across lamp	Approx. 18 V _{DC}

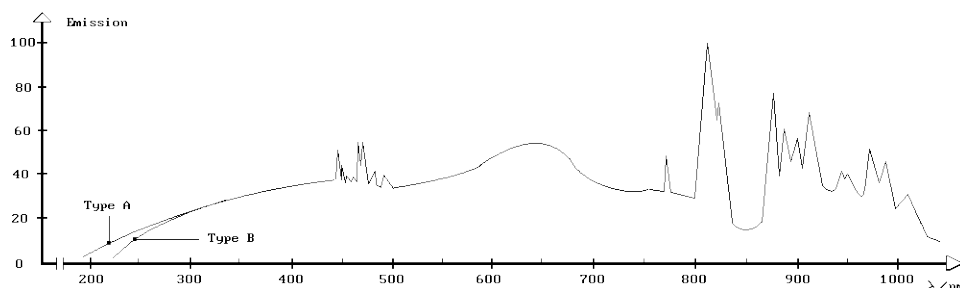


Figure 1.4. Spectral distribution for the two xenon lamp types.

1.4.1 ER130 fibre holder

The base plate for the fibre holder is soldered into the telescope. The translator mechanism, which is designed in the same way as for the lamp, is attached to the base plate with four screws behind the translators.

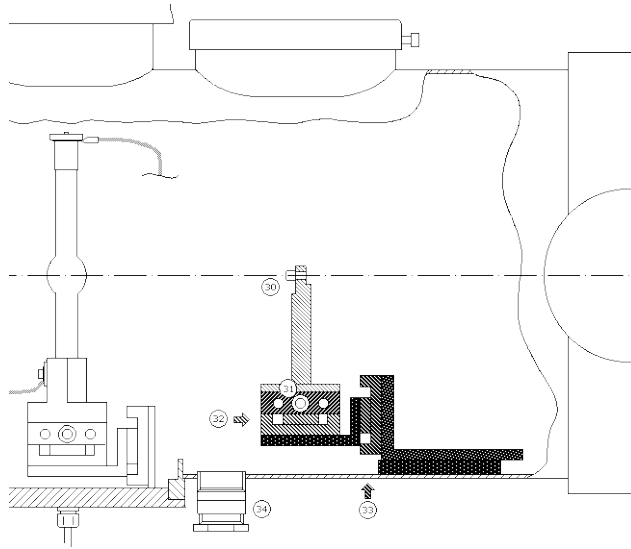


Figure 1.5. The fibre holder and its attachment inside the telescope.

ER130 fibre holder	
30	Threaded fibre-optic connector
31	Horizontal adjustment screw
32	Focusing screw
33	Access to vertical adjustment screw
34	Access for optical fibre and cable grip nut

1.5 The Retro-Reflector RR090

The retro-reflector unit RR090 consists of a number of corner cube prisms, which reflects the incident light beam straight back towards the transceiver. The prism housing is mounted on a ball monopod, which is lockable. The reflector unit is provided with an attachment for the mounting on a lamp post, or similar.

Theoretically, the incident angle is a less significant parameter, as the beam should always be reflected 180° . However, it has been shown that the reflected beam may be shifted 0.1 or 0.2 dm sideways over longer distances, when the reflector is mounted at a certain angle with the incident beam. This shift could cause a significant drop in the light level when monitoring.

The retro-reflector should therefore be mounted as linear as possible with the incident beam. The angle should not exceed 5° .

The monopod can be attached onto the prism housing on both the rear side and the underside. If the travel of the handle doesn't lock the monopod properly, it is possible to adjust the mechanism. Then unscrew the handle. The nut in which the handle was attached can then be tightened half a turn. Screw back the handle and check the function.

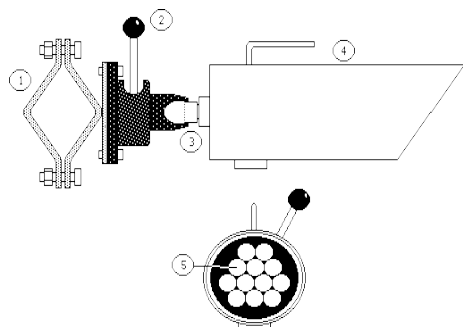


Figure 1.6. The retro reflector RR090.

RR090	
1	Lamp post attachment
2	Handle for locking the monopod
3	Monopod
4	Prism housing
5	Prisms (reflectors with 12 and 7 prisms are available)

1.6 The control unit

The control unit translates the signals which are delivered from the analyser into pulses for the two stepping motors on the ER130 transceiver. The unit contains two stepping motor card, one for each movement, a power supply card, and a controller card for the UV flip-flop.

The control unit can be supplied by either 115 or 230 V_{AC}. However, the fuse holder above the cable connection has to be turned to the appropriate position. The voltage setting is shown in the small window on the fuse holder. Changing from one setting to another is performed as follows:

- Pull out the fuse holder from the connection pad.
- Change the fuses according to the ratings below.
- Turn the holder so that the correct rating corresponds to the mark on the pad, see [figure 1.7](#). Push the holder back in position.

Fuses for the mains input	
230 V	250 V/2.5 A slow (2 pcs.)
115 V	250 V/5 A slow (1 pcs.)

Fuses are also found on the power supply card and the flip-flop power card. These cards are described in [section 1.6.1](#) and [section 1.6.3](#), respectively.

The connections to the analyser should be made to the ER130 interface (labeled F on the connection pad) and the multiplexer port (labeled H). The signals to the UV flip-flop are

supplied from the multiplexer interface. The cable is further described in section 1.6.4.

Note that the analyser has to be prepared for the operation of the ER130. Both the internal connections to the ER130 interface, and a few additional circuits on the I-86 co-processor card are required.

The two stepping motors are connected to the controller unit with 14-pins Cannon connectors. In order to distinguish between the horizontal and the vertical movement, one connector is female and the other one is male.

The UV flip-flop, finally, is connected with a 5 pin Cannon plug to the control unit.

When in operation the control unit should be kept in a dust free, temperature controlled environment. The working temperature range is 5 to 30 °C.

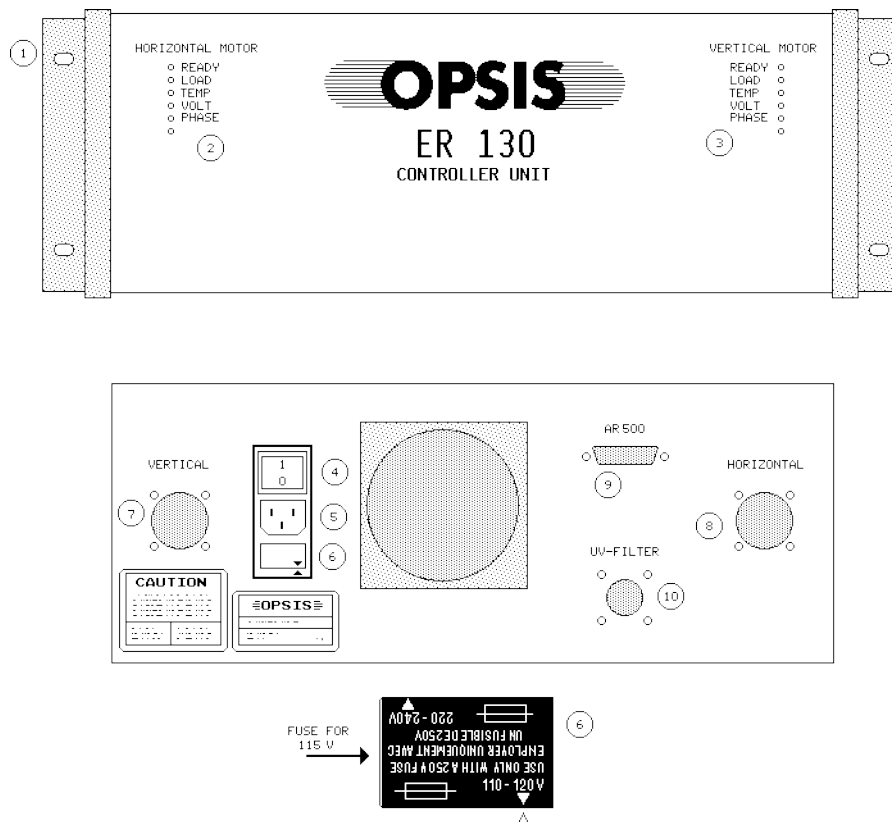


Figure 1.7. The control unit, the front and the rear side. All connections are made on the rear side.

ER130 control unit	
1	Attachment for rack mounting
2	Stepping motor indicators, horizontal movement
3	Stepping motor indicators, vertical movement
4	On/off switch
5	Mains input, 115 or 230 V _{AC}

ER130 control unit	
6	Fuse holder
7	Male Cannon connector, 14 pins, for the vertical stepping motor
8	Female Cannon connector, 14 pins, for the horizontal stepping motor
9	D-sub connector, 25 pins, for the analyser cable
10	Female Cannon connector, 5 pins, for the UV flip-flop

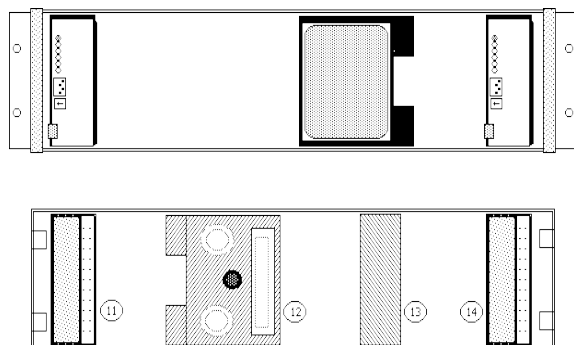


Figure 1.8. The ER130 control unit, where the rear plate is removed. (11): vertical stepping motor card, (12): power supply card, (13): flip-flop power card, (14): horizontal stepping motor card.

1.6.1 The power supply card (PM0)

The power module card, PM0, is transforming the input voltage of 230 or 115 V_{AC} to a voltage of 24 V_{DC}, which supplies the two stepping motor cards. The PM0 card is also used in the analyser.

The rating for the fuse on the secondary side of the transformer should be 250 V/8 A, slow.

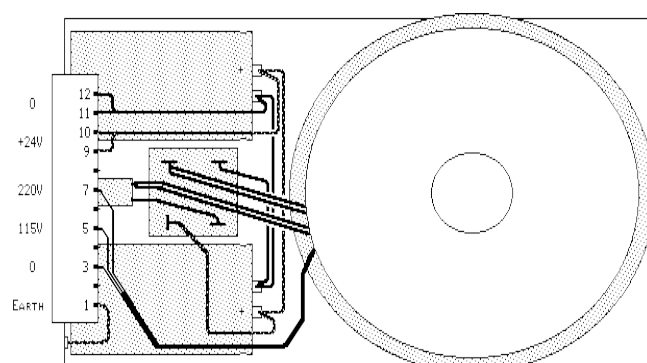


Figure 1.9. The power module card, PM0. The fuse is accessible from the rear side.

1.6.2 The stepping motor cards

The stepping motor cards are power drive units for driving the two 5-phase stepping motors on the ER130 unit. They establish the connection between the digital control in the analyser and the motors.

On the front side of the cards there are five red and a green light emitting diode, four toggle switches, and a rotary switch. The six diodes are visible through the front plate on the control box.

The toggle switches [7] are fundamental for the operation of the motor, and they should be set as indicated in the figure to the right.

The rotary switch [8] sets the phase current to the motor. The switch position D (“8 o’clock”) is the nominal setting for the VRGM 5913/50 stepping motor, which is currently being used. The switch should not be moved.

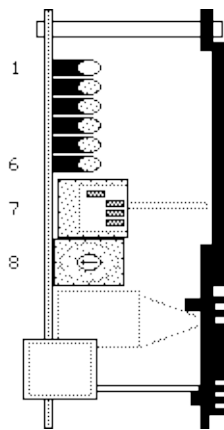


Figure 1.10. Front of the stepping motor card.

The six diodes, numbered 1 to 6 from the top, have the following indications:

1. Green diode, indicates readiness. The diode is on only in absence of malfunctions.
2. Red diode, short circuit indication. Lights up if the phase current has exceeded a value of 10 A for 2 μ s.
3. Red diode, excessive temperature. This is indicated if the heatsink temperature exceeds approximately 85 °C. If this diode lights up, check that the ventilation of the control unit is sufficient. Also check for short-circuits.
4. Red diode, under-/overvoltage. Lights up in case of operating voltages outside the permissible limits (lower than 24 V or higher than 35 V).
5. Red diode, phase interruption. This is indicated if there is an interruption in the motor circuit, for instance due to misconnected motor lines.
6. Red diode which has no function.

If any of the red diodes lights up, or if the green diode goes off, turn off the control unit and try to locate and rectify the fault before the unit is switched on again.

The flip-flop power card is designed for the operation of the optional UV flip-flop unit. The card is installed inside the ER130 control unit as a standard although the filter unit is optional, see [section 1.7](#).

The diagram illustrates the internal components of the power supply unit. Key components include:

- FUSE:** A 2.5A fuse (T 2.5A) is located at the top left.
- Transformer:** A large circular transformer is positioned in the center.
- Rectifier Bridge:** A bridge rectifier is shown on the right side.
- Output Filter:** Two electrolytic capacitors are connected to the output of the rectifier bridge, providing +12V and -12V outputs.
- Grounding:** The unit is grounded (GND) at the bottom left.

The fuse on the secondary side should have the rating T2.5A

The diagram illustrates the DSUB M/F-25p cable assembly. It consists of the following components:

- 1**: DSUB M-25p connector (male).
- 2**: DSUB F-25p connector (female).
- 3**: DSUB M-25p connector (male).
- 4**: Cable assembly connecting the two DSUB connectors.
- 5**: DSUB F-25p connector (female).

The cable assembly (4) is shown connecting the two DSUB connectors (1 and 2, and 3 and 5). The cable has a braided shield and a central conductor.

DSUB M-25p.

Pin.	Signal
1	+12V
3	-12V
14	Ver. STEP
15	Ver. DIR
18	Hor. STEP
19	Hor. DIR
22	Hor. POS
23	Ver. POS
24	UV-Filter
25	0V

DSUB F-25p.

Pin.	Signal
4	+12V
5	-12V
6	Ver. STEP
7	Ver. DIR
8	Hor. STEP
9	Hor. DIR
10	Hor. POS
11	Ver. POS
12	UV-Filter
13	0V

12

ER130 communication cable	
1	DSUB M-25p for connection to analyser port F.
2	DSUB F-9p for connection to analyser port H.
3	DSUB F-25p for connection to the ER130 control unit.
4	Cable length 0.25 m
5	Cable length 2 m (standard), max. 10 m.



Figure 1.13. The connections to the UV filter magnet. The pins D and E are used on the Cannon plug. The cable length should not exceed 25 metres.

The cable between the flip-flop and the control unit is a 2 wires cable, which is terminated with a 5 pins male Cannon connector. Figure 1.13 shows the internal connections to the control card, and the wiring configuration.

ER130 connections to UV filter magnet	
1	Cannon F-5p for connection to the ER130 control unit.
2	Cable length 5 m (standard), 25 m (max.)
3	Terminations to the UF 225 unit.

The stepping motor cables are terminated with Cannon connectors, one is male, the other one female, see figure 1.14 below.

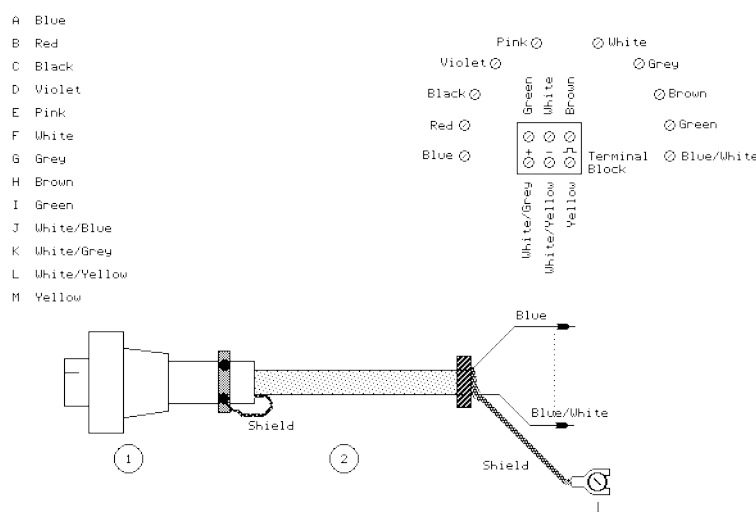


Figure 1.14. Wiring diagram for the standard cable between the stepping motor and the control unit, including the terminations to the stepping motor.

ER130 cable stepping motor/control unit	
1	Horizontal: Cannon Male 14p Vertical: Cannon Female 14p
2	Cable length 5 m (standard), see below

The maximum cable lengths are dependent on the cross-section of the wires. Due to the high frequency currents, the cables have to be screened and the wires should be twisted in pairs when using longer cables.

- The standard cable length is 5 metres. The cable is screened, but the wires are not twisted in pairs.
- With a wire cross-section of 0,75 mm² the cable length can be extended to max. 10 metres.

1.7 The UV Filter UF225

The UV filter flip flop UF225 is an optional unit which should be attached to the fibre post, as indicated in the figure below. The UF 225 consists of a filter holder and a magnet.

The UV should be inserted in front of the opto-fibre when nitric oxide and/or ammonia are being measured, in order to suppress the stray light level inside the spectrometer. Due to the lower light level in the ER130, compared with the ER110 and the ER150 units, the filter should not be fixed. It has to be removed when the other components are being measured. Therefore it is mounted on a flip-flop, which is controlled by the computer in the analyser. The operation is entered in the Installation menu, see the analyser software manual.

The flip-flop magnet has two positions. In the off position the filter is removed from the light beam. The magnet is connected to the flip-flop power card inside the ER130 control unit (see [section 1.6](#)).

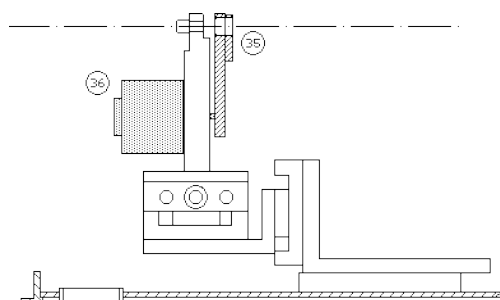


Figure 1.15. The UF225 unit.

UF225 unit	
35	filter holder
36	magnet

Before using the emitter/receiver 2

2.1 Safety regulations

- The xenon lamp in the emitter is a continuous discharge lamp which is initiated by a high-voltage pulse. The voltage of this pulse is about 30 000 volts. **Failure to follow the installation instructions and safety regulations may endanger life.**
- **Protective glasses shall be worn at all times when working with the lamp due to the risk of explosion.** The xenon lamp is a high-pressure lamp. This means that the pressure in the lamp at room temperature is about 10 bar, while during operation it rises to about 30 bar. A small scratch on the lamp may cause it to explode, propelling small fragments of glass with great force. It is therefore also recommended that long-sleeved clothing and gloves be worn, thus protecting the skin. Refrain from removing the protective sheath as far as is possible.
- **Never touch the glass with bare fingers.** Fingerprints reduce the transmission of the glass, which increases the temperature of the bulb, and thus the risk of explosion. Dust and fingerprints can be removed by carefully cleaning the lamp with a soft cotton cloth and pure alcohol or acetone.
- **The lamp must be installed with the anode (+) upwards.** It may lean at most 15 degrees from the vertical during operation. Incorrect installation of the lamp will lead to an alteration in its spectrum, resulting in errors in the measurements. There is also a considerable risk that the lamp will not function at all.
- **Ultraviolet radiation is emitted by the lamp.** Never look directly into the lamp without protective glasses. Naked skin should also be protected from UV irradiation.
- **During operation the lamp becomes very hot.** Do not, therefore, touch the lamp until it has cooled down.
- **Used lamps should be handled as carefully as new ones.** Never leave a lamp without a protective sheath. When changing a lamp, the protective sheath should be transferred from the new lamp to the old one.
- **When the lamp ignition unit is integrated inside the power supply PS150 unit (older versions only), make sure that the earth connection between the power supply and the ER110 emitter is properly connected before switching on power to the lamp.**

2.2 Precautionary measures

All service and maintenance inside the instruments may be done only by service personnel authorized by Opsis.

- The emitter and receiver should be stably mounted. They should be securely screwed down to avoid the influence of wind and weather. (Ensure that adequate safety measures have been taken for installation personnel working in areas where there is a risk of falling.)
- Use only the Opsis PS150 power supply for the xenon lamp.
- Ensure that the power supply is connected to a 230 V AC (+ 6, -10 %) or 115 V (± 10 %) mains supply, and that the supply unit is properly earthed.
- Never work on the power supply while it is connected to the mains.
- Ensure that the cables are correctly connected to the power supply before it is connected to the mains.
- Use only original cables and never make joins in cables.
- The optical fibre is very fragile. It should, therefore, not be subjected to blows or pressure, and should not be bent to a radius of less than about 10 cm. The fibre is also very sensitive to axial torsion. Do not, therefore, allow the fibre to hang freely from a height of greater than about 5 m.
- The polished ends of the fibre are very sensitive to dust and dirt. Ensure that they are protected at all times. If they should need to be cleaned, a soft cotton cloth or lens tissues may be used, with a little alcohol if necessary. Never touch the ends of the fibres with your fingers, and do not use fabrics containing synthetic fibres for cleaning.
- Never couple two fibre-optic cables without first ensuring that the ends are perfectly clean. In a threaded connection between two such cables, the ends of the fibres are no more than a few micrometres away from each other. Dust and dirt which may have found their way onto the connection can easily damage the ends of the fibres.
- Never touch the reflective surface of the two mirrors. The surface coating is very sensitive.

In choosing the site for the equipment, the following should be borne in mind.

- Try to minimize distance between the transceiver and the analyser. If possible, the fibre-optic cables should be no longer than 10 metres.
- A mains supply of 230 V_{AC} (or 115 V_{AC}) must be available for both the transceiver, the controller unit, and the analyser. A telephone connection should also be available at the analyser site.
- The analyser and the controller unit must be temperature controlled. Avoid exposing the units to dust, humidity, vibrations, etc.
- The ER130 unit must be stably mounted on a flat surface. When installing the unit on a table or a roof, make sure that access is given for the cables which are threaded through the base plate.
- Check that the telescope is allowed to move freely in all directions.
- Also check that the directions towards the retro-reflectors are within the working range of the telescope.
- The retro-reflectors should be directed towards the ER130 unit within an angle of 25°. The distance to the ER130 should not exceed 250 metres (150 metres when the 7 prisms retro is used).
- Ensure that the fibre optic cable is not crushed around any corners, and that it cannot be trampled to pieces.

When all units have been installed and properly secured, all cables between the ER130 unit, the controller unit, and the analyser can be connected. Make sure that the input voltage selection plugs are correct (115 V or 230 V).

- Switch on the lamp in the ER130.
- Switch on power to the controller unit. The green diodes on the front should light up.

The power to the analyser should never be switched on until the xenon lamp has turned on. In order to avoid problems in case of accidental power interruptions, the analyser mains supply should be provided with a time delay unit, set at approximately 15 minutes.

How to optimize the light and how to run the unit from the analyser will be described in the following sections.

3.1 Optimizing the light

The adjustments of the telescope is performed in two steps.

- When major operations on the telescope have been made, for instance when the rear mirror has been removed, the adjustments of all components have to be made from scratch. This procedure is treated in the section below.
- After for instance a lamp replacement only minor adjustments are necessary. The procedure is described in [section 3.1.2](#).

3.1.1 Starting from scratch

The procedure is performed in two steps:

- alignment of the beam along the centre axis of the telescope towards a reflector.
- without moving the lamp or the lamp mirror, adjusting the receiving mirror so that the reflected beam can be focused onto the opto-fibre.

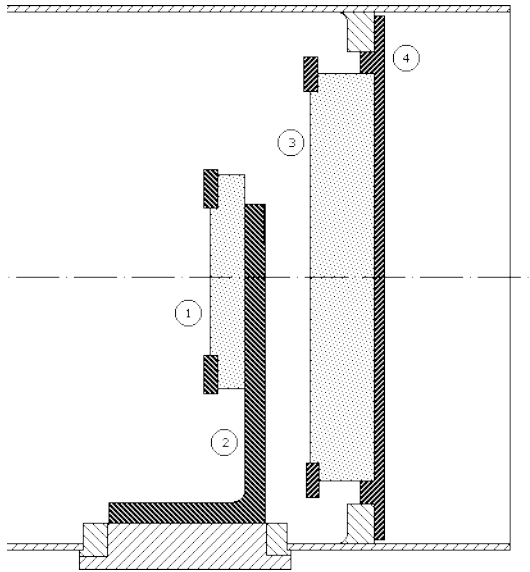


Figure 3.1. The attachments of the two mirrors in the telescope. The small mirror is mounted on a holder from below, while the big mirror is attached from behind. To gain access to the big mirror the rear plate of the telescope has to be removed. How the two mirrors are adjustable is explained in the next figure. (1): small mirror, diameter 76 mm, (2): mirror holder, (3): bigger mirror, diameter 150 mm, (4): mirror attachment.

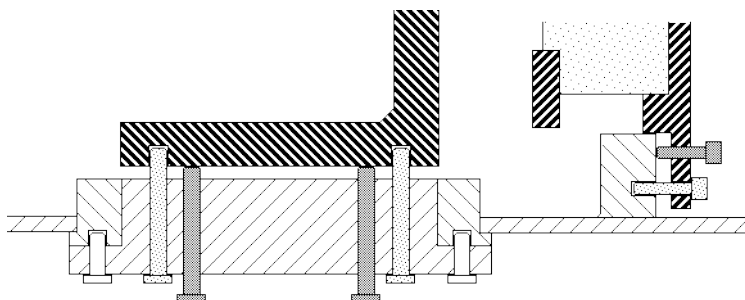


Figure 3.2. The mirror adjustment screws. See the text below for explanations.

The big mirror holder is attached to the telescope by three screws (light grey in the figure). To make it possible to adjust the angle of the mirror holder, three screws are threaded through the holder (dark grey).

The smaller mirror is adjusted from below. By loosening the four screws (white in the figure) it is possible to rotate the holder slightly. This corresponds to a horizontal adjustment. The vertical adjustment is performed in a similar way as for the bigger mirror. The two dark grey screws are pushing, while the four light grey ones are pulling.

Always start the mirror adjustments by removing the dark grey “pushing” screws entirely. When the adjustments are completed, the holder should not be resting on the pushing screws only. This means that when the light grey pulling screws are tightened at least one pushing screw on each mirror should still be loose.

NOTE: Before starting any adjustments, study the safety regulations in section 2 carefully. Always wear protection glasses, not only for the mechanical protection of the eyes, but also for filtering the strong UV radiation.

Required tools for the adjustments are:

- a set of Allen keys
- an optical fibre, 1 to 2 metres long
- a lux meter LM010.

Performance

1. Switch on the lamp, remove the front window, and aim the telescope at a reflection screen, or a similar device where the light spot is clearly visible, placed at least 50 metres away. In order to lock the telescope in the desired direction the control unit to the stepping motors can be switched on.
2. Move all six translators to their middle positions (three for the lamp, three for the fibre).
3. Adjust the lamp mirror so that the beam is aligned with the centre axis of the telescope towards the screen. Tighten the screws to the mirror.

4. Adjust on the focusing screw on the lamp so that the image on the screen is as small and bright as possible. The translator should now be 1 or 2 mm from its front position, as the distance to the screen is fairly short. If the translator movement is not sufficient, and the distance is more than 50 metres, it may be necessary to insert an extension ring between the telescope and the mirror holder, in order to move the mirror backwards. The thickness of the extension ring have to be tried out, but should not be more than 5 mm. Remove the mirror holder, insert the ring and repeat the steps 1 to 4 above.
5. Replace the screen by a retro-reflector, and put it in the brightest part of the image.
6. Use a piece of paper to catch the reflected spot from the receiving mirror. Adjust the receiving mirror so that the spot is centred in the fibre connector. Tighten all screws on the mirror holder. The smallest spot should be found just inside the fibre holder.

The mirrors should from now on not be moved. All adjustments should be made on the translators only. The transceiver is ready for the fine adjustments of the light level.

3.1.2 Precision adjustments

7. Switch off the lamp and let it cool down. Mount the opto-fibre in the holder and switch on the lamp again. Allow the lamp to warm up for 10 minutes.
8. Aim the telescope at a retro-reflector, placed approximately 100 metres away from the transceiver. Lock the telescope in the position by switching on the stepping motor control unit.
9. Connect the lux meter to the opto-fibre. Check that the most sensitive range is active.
10. If the lux meter reading is zero, use a thin piece of paper to check the position of the reflected spot in front of the fibre. Adjust the fibre holder vertically and horizontally until the meter is registering.
11. Optimize the lateral and vertical adjustments for the fibre holder step by step.
12. Optimize the lamp laterally and vertically step by step.
13. Adjust the focusing screw for the fibre half a turn. Optimize the fibre holder laterally and vertically. If the light level went down, move the focusing screw the other direction. If the light level was improved, repeat the focusing adjustments with half a turn between each optimization until no improvement is registered.
14. Repeat step 13 for the lamp. Optimize both the fibre and the lamp laterally and vertically between each focusing adjustment.

Note that for each adjustment of lamp the fibre has to be re-adjusted, as the whole beam-configuration in the telescope changes when the lamp is moved.

The final lux reading should be in the order of 100 lux on a distance of 100 metres, and about 70 lux when the retro-reflector is 150 metres away.

Small adjustments may be necessary from time to time. Factors which can affect the light intensity reaching the analyser are:

- movement of the lamp holder and mirror in the transmitter caused by heat generated by the lamp,
- instability of the transmitter mounting. Metal roofing will, for example, expand when heated by the sun in the summer.

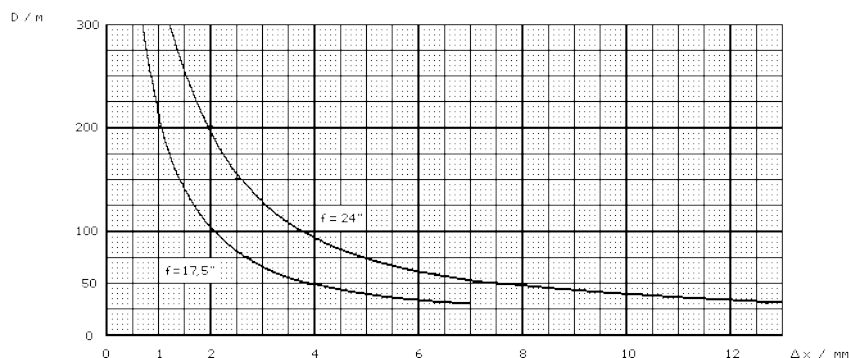


Figure 3.3. The required axial adjustment Δx as a function of the distance D to the retro-reflector for the two mirrors in the transceiver. See the example below.

Example (referring to figure 3.3): If the transceiver was optimized when the reflector was placed at a distance of 50 meters, and then moved to 200 metres, the required focusing adjustment for the lamp is about $(7.5 - 2.0) \text{ mm} = 5.5 \text{ mm}$, while the fibre has to be moved only $(4.0 - 1.1) \text{ mm} = 2.9 \text{ mm}$. The movements should be towards the mirrors, as the reflector is moved away from the transceiver.

3.2 Installation procedure

When the light level in the transceiver has been optimized, and all units in the system are installed, it is time to store the coordinates for the directions of the reflectors in the analyser software. The procedure is described in detail in the Analyser Reference Manual. Below a short step by step instruction is given.

- Switch on the lamp in the ER130, and allow it to warmup at least 10 minutes.
- Switch on the analyser, allow it to warm up, and proceed into the ER130 installation menu. Make sure that the fibre optic cable is connected to the analyser.
- Without switching on the power to the controller unit, aim the telescope at the first retro-reflector.
- By looking along the telescope from behind it is possible to see the reflection from the reflector when the beam hits the target. Note that the reflected beam is very narrow, and it is thus necessary to look very close to the telescope.
- When a reflection is visible, switch on the power to the controller unit.
- Press [F5] Optimize in the installation menu. The ER130 will then scan over an area around the first direction in order to find the best light level.

- Press [F8] Seek path and enter the current path number. The ER130 then moves to its Home position, counting the number of steps, and finally stores the coordinates for the current path. The installation for the path is ready.

As long as the ER130 is not moved, the coordinates for the path will remain constant. However, in case of power interruptions the unit may have to be reset to its Home position before the correct directions are found again. This is done by pressing [F9] Reset ER130 in the installation menu.

Once the positions for the transceiver and the retro-reflectors are decided, the distances between them have to be measured as accurately as possible. These distance should then be entered in the Installation menu, see the analyser software manual.

NOTE: *The accuracy of the distance directly affects the accuracy of the results. A distance which is measured to be 10 % too short will lead to results that are 10 % too high.*

To measure the distance, use a measuring tape, or, when the distance is longer, a detailed map over the area. The map should have a scale 1: 10 000, or better.

4.1 Cleaning the mirrors and windows

The protective windows are made of quartz, and cleaning, therefore, causes no problems. Dirt and grease can be removed using a soft cotton cloth with acetone or pure alcohol. If the inside of the window requires cleaning, it should first be removed from the telescope, cleaned, and allowed to dry completely before being remounted. This will prevent unnecessary moisture and solvent vapour from getting into the telescope.

The mirrors are made of a thin layer of aluminium which is sputtered onto a quartz substrate. To protect the highly reflective aluminium surface from dirt and oxidation, and to obtain a high reflectance in the short-UV wavelength region, a layer of magnesium fluoride, MgF_2 , has been deposited on the mirrors in the ER110.

The magnesium fluoride coating on the mirrors is very sensitive. It must not be touched or cleaned with liquids.

Magnesium fluoride is a salt which is soluble in most liquids, including water. Therefore, any dirt finding its way onto the mirrors must remain there. This will normally not pose a problem, as the mirrors are located far back in the telescopes. Loose dust may be removed gently with a dry, soft, cotton cloth or a photographer's lens brush.

The reflective surface is affected by the moisture which is present in the atmosphere. In time, the surface may become somewhat greyish, and this will affect the amount of light reaching the analyser. This process takes place more rapidly in the receiver than in the transmitter as the mirror in the transmitter is, to some extent, protected by the heat given off by the lamp. After a few years, it may, therefore, be necessary to replace one or both of the mirrors.

4.2 Lamp replacement

The lamp and its holder are shown in [figure 1.3](#). Great care must be taken when mounting or removing the lamp. Before attempting to mount or remove a lamp, the following must be observed.

- Read through the safety regulations in [section 2.1](#).
- Ensure that the power supply is disconnected from the mains and then disconnect the lamp from the power supply.

- Use protective glasses. Cover naked skin on hands and arms.
- Do not remove the protective sheath from the new lamp until absolutely necessary.
- Make sure that the lamp has cooled down entirely.
- Never touch the glass. Fingerprints will be burnt onto the glass raising the operating temperature and increasing the risk of explosion.

4.2.1 Procedure for lamp replacement

1. Unscrew the window [21] from the telescope. Unscrew the nut holding the lamp power cable [23]. Remove the lamp from its socket, after having loosened the two screws [25] holding it in place.
2. The new lamp is mounted in the same way as the old one. Start by transfer-ring the protective sheath to the old lamp. Be careful to mount the new lamp with the anode, marked (+), upwards. Ensure that the lamp is pushed well down into the socket. Tighten screws [25] carefully, do not tighten them too much.
3. The white teflon-insulated cable is connected to the anode (positive terminal) on the top of the lamp. The nut [23] is then tightened well but carefully. If the cable shows signs of oxidation it should be cleaned before being reconnected. Ensure that the teflon insulation is not trapped under the nut.
4. Remount the window in the telescope before the lamp is turned on.

If the lamp has been remounted correctly, the transceiver optimization will not have changed significantly. However, some fine lateral or vertical adjustment, using the translator mechanisms, may be necessary (see [section 3.1.2](#)).

This section provides guidance for determining the causes of malfunctions or anomalies during operation. The guidance is given in the form of a troubleshooting scheme, which may help to diagnose the problems.

Problem	Possible causes and action
The lamp does not function.	Check mains supply to power supply Check that power supply is properly earthed. Check that power cables are correctly connected and grip nuts tightened. Check that the power cable to the lamp anode is correctly connected and is not oxidized. Check that the lamp is properly earthed. Check the fuses in the power supply. See the manual for the PS150 unit. The lamp may be burnt out. Replace with a new one. If the lamp still does not work, the power supply is probably faulty. Contact your Opsis representative.
There is no light reaching the receiver.	The lamp has gone. Bad visibility, preventing the light from reaching the receiver. The transceiver is poorly aligned. The transceiver has lost the directions. Check the stability of the foundation. Also check the cables between the units so that bad connections can be excluded.
Light is reaching the receiver but the fibre is transmitting no signal.	The receiver/fibre holder is poorly adjusted. Check where the light spot is located. The optical fibre is broken. The optical fibre is badly connected.
Poor light is indicated for some substances being measured.	The lamp may be worn. The reflective layer on the mirror may be damaged.

